```
// ==========================================================
// KOTLIN KICKSTART LAB
// Topic: Your First Kotlin Program – Variables & Functions
// Duration: ~40–50 minutes
// ==========================================================
//
// 🧠 Overview:
// In this lab, you will build your first complete Kotlin program
// that presents your learning summary like a session report.
// You'll greet yourself as the learner, show progress using variables,
// use functions for calculation, and summarize what you learned.
//
// Follow the comments carefully – each part has HINTS and SAMPLE OUTPUTS.
// ----------------------------------------------------------

fun main() {

    println("========== Kotlin Kickstart: Session Progress Report ==========\n")

    // ----------------------------------------------------
    // PART 1: Introducing Yourself
    // ----------------------------------------------------
    // TODO 1: Create a variable (val) to store your name.
    // Example: val userName = "Aditi"
    //
    // TODO 2: Print a short introduction line that sounds like a report.
    // Hint: Use string templates —
    // println("Hello, I'm $userName, and this is my Kotlin learning report!")
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//
// SAMPLE OUTPUT:
// Hello, I'm Aditi, and this is my Kotlin learning report for today.
// -----------------------------------------------------
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// ----------------------------------------------------
// PART 2: Sharing Your Practice Stats (val vs var)
// ----------------------------------------------------
// Imagine you're tracking your learning progress:
// val → fixed data (initial goal)
// var → flexible data (current progress)
//
// TODO 1: Declare val initialScore = 10
// TODO 2: Declare var currentScore = 10
// TODO 3: Increase currentScore by 15 and print a report line.
//
// SAMPLE OUTPUT:
// My initial goal score was 10, but now I've reached 25 points!
// ----------------------------------------------------
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// ----------------------------------------------------
// PART 3: Reporting Learning Progress (Functions)
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// ----------------------------------------------------
// Define a function named calculateProgress(base: Int, bonus: Int): Int
// below main() (template already given below).
//
// TODO: Call your function here with any two integers and store the result
// in a variable named totalProgress.
// Then print: "In total, I've earned <totalProgress> learning points today."
//
// HINT: val totalProgress = calculateProgress(30, 20)
//
// SAMPLE OUTPUT:
// In total, I've earned 50 learning points today.
// ----------------------------------------------------
```

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// ----------------------------------------------------
// PART 4: Listing Topics You Explored
// ----------------------------------------------------
// You'll now list the topics you covered in this session.
//
// TODO 1: Create a MutableList<String> containing three topics you learned
// (e.g., "Variables", "Functions", "Strings").
//
// TODO 2: Add one more topic to the list.
// MINI HINT: list.add("TopicName")
//
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// TODO 3: Remove one topic from the list.
// MINI HINT: list.removeAt(index)
//
// TODO 4: Pass the list to summarizeTopics(list) function and print:
// "Today, I explored the following topics: <topics>"
//
// SAMPLE OUTPUT:
// Today, I explored the following topics: Variables, Strings, Loops
// -----------------------------------------------------
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// ----------------------------------------------------
// PART 5 (Optional Challenge – Reflective Summary)
// ----------------------------------------------------
// Create and call a function encourageUser(name: String, topics: List<String>)
// that prints:
// "I'm proud of myself! I, <name>, completed <count> topics in this session."
//
// MINI HINT:
// Use topics.size and string templates.
// Example:
// println("I'm proud of myself! I, $name, completed ${topics.size} topics in this session.")
//
// SAMPLE OUTPUT:
// I'm proud of myself! I, Aditi, completed 3 topics in this session.
```

```
    // ------------------------------------------------------

    println("\n========== End of My Kotlin Learning Report ==========")
}

// ==========================================================
// FUNCTION SECTION – define your functions below main()
// ==========================================================

// FUNCTION 1: calculateProgress
// Purpose: Accepts two Int values (base and bonus) and returns their sum.
// ------------------------------------------------------
fun calculateProgress(base: Int, bonus: Int): Int {
    return base + bonus
}

// FUNCTION 2: summarizeTopics
// Purpose: Takes a List<String> and returns them joined with commas.
// ------------------------------------------------------
fun summarizeTopics(topics: List<String>): String {
    return topics.joinToString(", ")
}

// FUNCTION 3 (Optional): encourageUser
// Purpose: Print a reflective summary line with the user's name and topics count.
// ------------------------------------------------------
fun encourageUser(name: String, topics: List<String>) {
```

```
    println("I'm proud of myself! I, $name, completed ${topics.size} topics in this
session.")
}

// ============================================================
// SAMPLE OUTPUT
// ============================================================
//
// ========== Kotlin Kickstart: Session Progress Report ==========
//
// Hello, I'm Aditi, and this is my Kotlin learning report for today.
//
// My initial goal score was 10, but now I've reached 25 points!
//
// In total, I've earned 50 learning points today.
//
// Today, I explored the following topics: Variables, Strings, Loops
//
// I'm proud of myself! I, Aditi, completed 3 topics in this session.
//
// ========== End of My Kotlin Learning Report ==========
```

```
fun main() {

    println("========== Kotlin Kickstart: Session Progress Report ==========\n")
    val userName = "Sanjana"
    println("Hello, I'm $userName, and this is my Kotlin learning report!")

    val initialScore = 10
    var currentScore = 10
    currentScore += 15

    println("My initial goal score was $initialScore, but now I've reached $currentScore
points!\n")

    val totalProgress = calculateProgress(30, 20)
    println("In total, I've earned $totalProgress learning points today.\n")

    val list = mutableListOf("Variables", "Functions", "Strings")
    list.add("Loops")
    list.removeAt(1)

    val listSummary = summarizeList(list)
    println("Today, I explored the following topics: $listSummary\n")

    encourageUser(userName, list)

    println("\n========== End of My Kotlin Learning Report ==========")
}
```

```
fun calculateProgress(base: Int, bonus: Int): Int {
    return base + bonus
}

fun summarizeList(list: List<String>): String {
    return list.joinToString(", ")
}

fun encourageUser(name: String, topics: List<String>) {
    println("I'm proud of myself! I, $name, completed ${topics.size} topics in this session.")
}
```